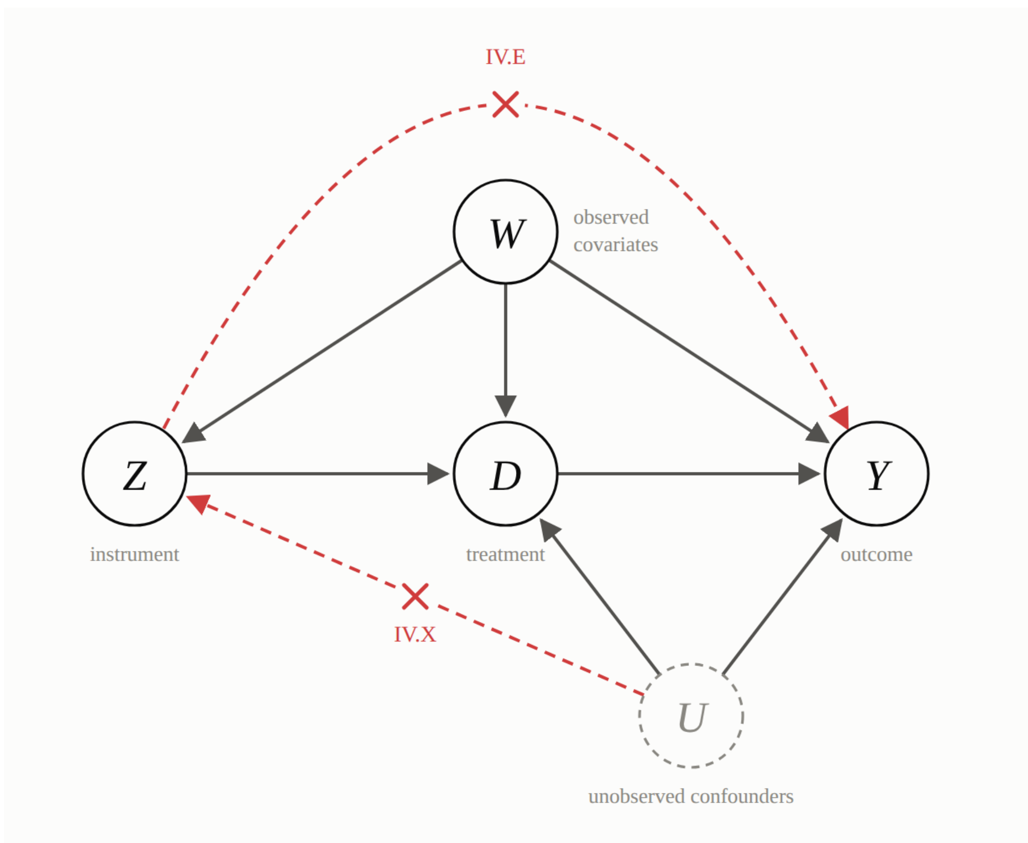
1 Instrumental Variables

This short note provides a general treatment towards instrumental variables.

In words, an instrument Z must be:

1. **Relevant**. It is correlated with the treatment variable D.
2. **Excludable**. It has no *direct* effect on the outcome variable Y.
3. **Exogenous**. It is uncorrelated with all variables U that are correlated with the outcome other than the treatment variable, possibly conditioning on covariates.



Let the observed data be $\mathcal{V} = (Y,D,W,Z)$. Y is the outcome, D is the treatment, W is the covariates, Z is the instrument.

We are interested in a true parameter θ_0 and know a residual function $\rho : \mathcal{V} \times \Theta \to \mathbb{R}^{d_p}$. Note that $\theta \in \Theta$ may be infinite-dimensional. Define a function on the instruments, $f : \mathrm{supp}(Z, W) \to \mathbb{R}^{d_f \times d_p}$. The model is identified by the single moment condition

$$\mathbb{E}[f(Z, W)\rho(V; \theta_0)] = \mathbf{0}_{d_f}$$

which follows from the assumptions below.

1. **Exogeneity** and **Exclusion**.

$$\mathbb{E}[\rho(V; \theta_0) \mid Z, W] = \mathbf{0}_{d_p}.$$

2. **Relevance**. $\mathbb{E}[f(Z, W)\rho(V; \theta)] = \mathbf{0}_{d_f} \quad \forall f \in \mathcal{F}$ if and only if $\theta = \theta_0$. Here $\mathcal{F}$ is a class of functions, and can be infinite dimensional

To see that these assumptions lead to the moment condition, simply write

$$\begin{aligned}
\mathbb{E}[f(Z,W)\rho(V;\theta_0)] &= \mathbb{E}\Big[\mathbb{E}\big[f(Z,W)\rho(V;\theta_0) \mid Z, W\big]\Big] \\
&= \mathbb{E}\Big[f(Z,W)\ \underbrace{\mathbb{E}[\rho(V;\theta_0) \mid Z, W]}_{=\mathbf{0}_{d_\rho}\text{ (by Assumption 1)}}\ \Big] \\
&= \mathbf{0}_{d_f}
\end{aligned}$$

1.1 NPIV Example

Consider the nonparametric IV setup,

$$Y = g_0(D, W) + \epsilon$$

where Y is the outcome, D is the endogenous treatment, W are the covariates, and Z are the instruments. The unknown parameter of interest is the infinite-dimensional function g_0. The residual function is given by $\rho(V; g) := Y - g(D, W)$.

- Here, the exogeneity and exclusion restriction is

$$\mathbb{E}[\rho(V; g_0)|Z, W] := \mathbb{E}[Y - g_0(D, W)|Z, W] = 0.$$

- And the relevance restriction is

$$\mathbb{E}[Y - g(D, W) \mid Z, W] = 0 \iff g = g_0.$$

Together, these two assumptions give the following. For any measurable instrument function $f(Z,W)$, we have

$$\begin{aligned}
\mathbb{E}[f(Z,W)(Y - g_0(D,W))] &= \mathbb{E}\Big[\mathbb{E}\big[f(Z,W)(Y - g_0(D,W)) \mid Z,W\big]\Big] \\
&= \mathbb{E}\Big[f(Z,W)\, \underbrace{\mathbb{E}[Y - g_0(D,W) \mid Z,W]}_{=0}\Big] \\
&= 0
\end{aligned}$$

While the derivation of the moment condition does not require the relevance restriction, it is required for g_0 to be identified. Since g_0 is infinite dimensional, identification requires this conditional restriction to hold over a rich class of test functions such as those spanning $L^2(Z,W)$.